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13 March 1972
SUPERSEDING
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17 December 1964

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 6AN8A

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Triode-pentode, miniature, medium Mu, sharp cutoff

Outline --- 6-2 (EIA)

Base --- E9-1

Envelope --- T6-1/2

Cathode --- Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8	9
Element	---	2a	2g	2k	h	h	1a	1g2	1g1	1k, 1g3, int sd

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	E1b	E1c1	E1cc2	E2b	E2c	P1a	P1g2	P2a	Ehk	Rg1	TE	Alt
Unit:	V	Vdc	Vdc	Vdc	Vdc	Vdc	W	W	W	v	Ohms	°C	ft
Maximum:	6.9	330	0	330	330	0	2.3	0.55	2.8	200	Note 2	140	Note 3
Minimum:	5.7	---	---	---	---	---	---	---	---	--- (Note 1)	---	---	---

TEST CONDITIONS:

Pentode (Sec 1)	6.3	125	-1.0	125	---	---	---	---	---	---	---	---	---
Triode (Sec 2)	6.3	---	---	---	150	-3.0	---	---	---	---	---	---	---

GENERAL:

Qualification - Required

Ⓐ denotes changes

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
1266	Total grid current	E1b = 200 Vdc; E2b = 135 Vdc; E1c2 = 150 Vdc; E1c1 = E2c = -2 Vdc (see notes 4 and 5)	0.65	II	Ic	0	-1.5	µAdc
1256	Electrode current (1), anode (Triode)		0.65	II	Ib	9	21	mAdc
1306	Transconductance (1) (Triode)		0.65	II	Sm	3,500	5,500	µmhos
1256	Electrode current (1), anode (Pentode)		0.65	II	Ib	7.2	16.8	mAdc
1306	Transconductance (1) (Pentode)		0.65	II	Sm	5,500	10,500	µmhos
1201	Short and discontinuity detection		0.4	II	---	---	---	---
	Quality conformance inspection, part 2							
1301	Heater current		4.0	S3	If	410	490	mA
1336	Heater-cathode leakage	See note 6	4.0	II	Ihk	---	20	µAdc
---	Pentode knee	See note 7	4.0	II	---	---	65	V
1306	Transconductance (2) (Triode)	Ef = 5.7 V	4.0	S3	Sm Ef	3,100	---	µmhos
1306	Transconductance (2) (Pentode)	Ef = 5.7 V	4.0	S3	Sm Ef	5,100	---	µmhos
1256	Electrode current (2), anode (Pentode)	Ec1 = -7.5 Vdc	4.0	S3	Ib	---	30	µAdc
1256	Electrode current, screen (Pentode)		4.0	S3	Ic2	1.4	5.2	mAdc
1256	Electrode current (2), anode (Triode)	Ec = -21 Vdc	4.0	II	Ib	---	100	µAdc
1316	Amplification factor (Triode)		4.0	S3	Mu	17	25	---
(A) 1246	Audio frequency noise	Ecal = 700 mVac (see note 8)	2.5	II	EB	---	17	vu
1211	Insulation of electrodes	See note 9	4.0	S3	R	100	---	Meg
1331	Direct-interelectrode capacitance (Pentode)	Without external shield	6.5	Code E	Cg1p Cin Cout	--- 5.5 1.8	0.047 8.5 3.0	pF pF pF
1331	Direct-interelectrode capacitance (Triode)	Without external shield	6.5	Code E	Cgp Cin Cout	1.1 1.5 1.1	1.9 2.5 1.9	pF pF pF
1369	Heater warmup time		---	---	t	6.8	15	sec

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 2 -Continued							
1121	Base strain		---	---	---	---	---	---
2126	Glass strain		2.5	I	---	---	---	---
1105	Permanence of marking		---	---	---	---	---	---
	Quality conformance inspection, part 3							
1501	Intermittent life	Group A; E1b = 200 Vdc; E1c1 = 0; E1c2 = 125 Vdc; R1k = 180 ohms; R1g1 = 1.0 Meg; E2b = 200 Vdc; E2c = 0; R2k = 560 ohms; R2g = 1.0 Meg; Ehk/k = +200 Vdc	---	---	---	---	---	---
---	Intermittent life-test end points (500 hours)	Transconductance (1) (Pentode)	---	---	Sm	4,900	---	µmhos
		Transconductance (1) (Triode)	---	---	Sm	2,900	---	µmhos
		Total grid current	---	---	Ic	0	-4.5	µAdc

NOTES:

1. The positive dc component of Ehk shall not exceed 100 V.
2. Rating values for external grid No. 1 circuit resistance are:

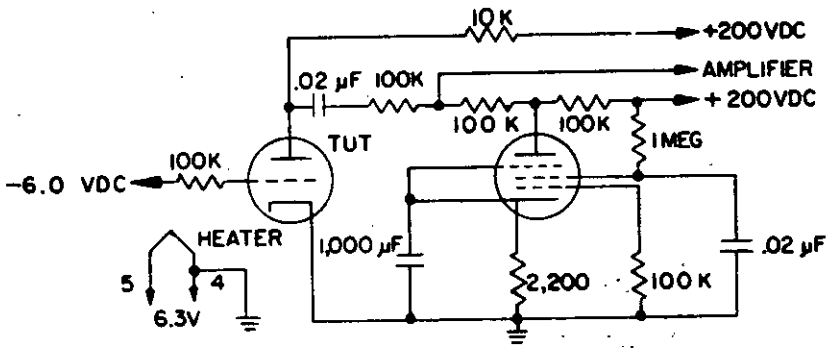
	Fixed bias	Cathode bias
Pentode	0.25 Meg	1.0 Meg
Triode	0.5 Meg	1.0 Meg

3. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.
4. Combined current shall be read for both units.
5. This test to be performed at the conclusion of the holding period.
6. Test with cathodes tied together.
7. This test is to be performed by displaying on a curve tracer an anode family (ib versus eb). With Ebb varied from 0 to 200 Vdc; R1 = 300 ohms; Ec2 = 150 Vdc, the knee of the Ec1 = 0 trace shall be determined as the point at which the trace is tangent to a straight line having a slope of 3.5 ma per 20 volts.

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NOTES: -Continued

(A) 8. Test in the following circuit:



9. Measure each tube section (or unit) separately.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Air Force - 17
DSA - ES

User activities:
Army - WC
Navy - AS, MC, CG, SH

Preparing activity: Army - EL

Agent: DSA - ES

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